

Lecture 7: State-space form, Linearization

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English version of slides by ©Bo Wahlberg



Today's Ordlista

English	Svenska
state-space (form/representation)	tillstånds(form/beskrivning)
linearization	linjärisering
state vector	tillståndsvektorn
Diagonal form	Diagonalform
Controllable canonical form	Styrbar kanonisk form
Observable canonical form	Observerbar kanonisk form
Fixed point	Stationär punkt
eigenvalues	egenvärden
Nonlinear	Olinjärt



Outline

Recap

State-space Representation

Linearization

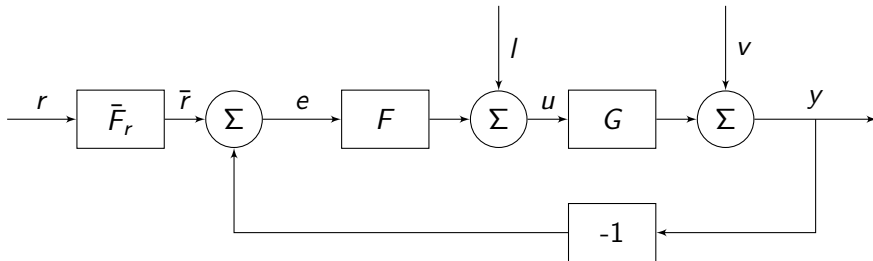
In the previous lecture..

- Sensitivity and robustness

Today:

- State-space representation
- Linearization

Recap: Sensitivity and Robustness



$$Y = \underbrace{\frac{GF}{1+GF}}_{G_c} \bar{R} + \underbrace{\frac{1}{1+GF}}_S V + \underbrace{\frac{G}{1+GF}}_{G_{ly}} L$$

$$U = \underbrace{\frac{F}{1+GF}}_{G_{ru}} (\bar{R} - V) + \underbrace{\frac{1}{1+GF}}_S L$$

Control Design $\Rightarrow$ Determine F ($G_o = GF$)

Recap: Four Tests

1. Plot $|G_c(i\omega)|$ (in particular, ω_B och M_p) [Fig. 5.8 in the book]

Goal: $G_c(i\omega) \approx 1 \quad \omega < \omega_B$

2. Plot $|S(i\omega)|$ (sensitivity function)

$$S(i\omega) + G_c(i\omega) = 1 \Rightarrow$$

$$S(i\omega) \approx 0 \quad \omega < \omega_B$$

Information on how **output**-disturbance is suppressed.

3. Plot $|G_{ly}(i\omega)|$

$$G_{ly} = \frac{G}{1+GF} \frac{F}{F} = G_c \frac{1}{F} \approx \frac{1}{F} \quad \omega < \omega_B$$

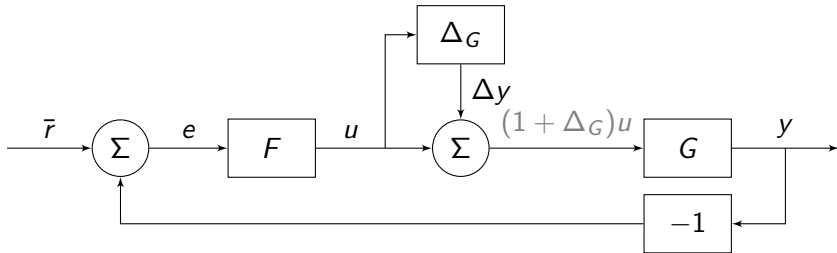
Information on how **input**-disturbance/noise is suppressed

4. Plot $|G_{ru}(i\omega)|$

$$G_{ru} = \frac{F}{1+GF} \frac{G}{G} = G_c \frac{1}{G} \approx \frac{1}{G} \quad \omega < \omega_B$$

Information about the magnitude of the control signal

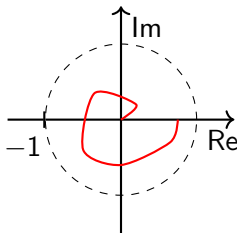
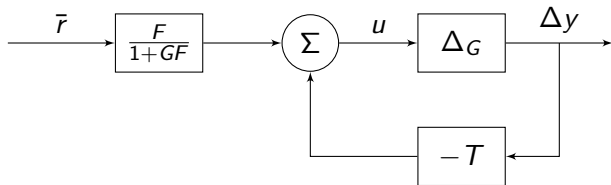
Recap: Robustness



$$\begin{cases} U &= -\frac{GF}{1+GF}\Delta Y + \frac{F}{1+GF}\bar{R} \\ \Delta Y &= \Delta_G U \\ T &= G_c = \frac{GF}{1+GF} \end{cases}$$

Recap: Robustness

Draw:



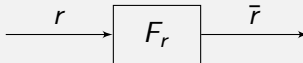
Sufficient condition: Stable if the open-loop gain is $< 1 \implies |T(i\omega)\Delta_G(i\omega)| < 1$

$$\implies |T(i\omega)| < \frac{1}{|\Delta_G(i\omega)|} \quad \forall \omega$$

i.e., we obtain the robustness criterion from F6!

Recap: Robustness

Example (Reference signal filtering)



F_r does **not** affect robustness or sensitivity

F_r affects servo characteristics/specifications in performance (i.e., $r \rightarrow y$)

For example, F_r can be a low-pass filter that provides smoother behavior for steps, etc.

Homework: Check in MATLAB the magnitude (e.g., $u_{\max}$) of the corresponding signal u when we apply a low-pass filter F_r vs when we apply directly a step r .



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Motivation for state-space representation

Differential equation:

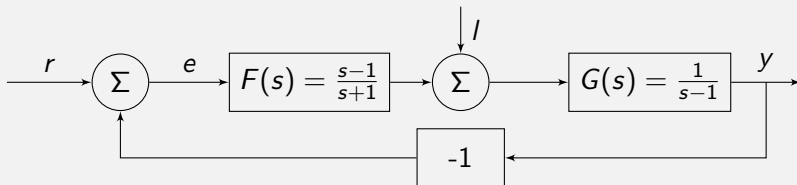
$$\ddot{y}(t) + \bar{a}_1 \dot{y}(t) + \bar{a}_2 y(t) = \bar{b}_0 \dot{u}(t) + \bar{b}_1 u(t)$$

Transfer function:

$$G(s) = \frac{\bar{b}_0 s + \bar{b}_1}{s^2 + \bar{a}_1 s + \bar{a}_2}$$

The problem with transfer functions

Example



$$G_c = \frac{GF}{1+GF} = \frac{\frac{s-1}{s+1} \frac{1}{s-1}}{1 + \frac{s-1}{s+1} \frac{1}{s-1}} = \frac{1}{s+2} \quad \text{Stable!}$$

$$G_{ly} = \frac{G}{1+GF} = \frac{\frac{1}{s-1}}{1 + \frac{s-1}{s+1} \frac{1}{s-1}} = \frac{s+1}{(s+2)(s-1)} \quad \text{Unstable!}$$

Warning:

Transfer function algebra can give “misleading” conclusions (due to initial values = 0).

The lesson is: One should not cancel out (hidden) unstable poles!

State-space representation

State-space model:

A 2nd order differential equation can be converted into two coupled 1st-order differential equations:

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \underbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}_A \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \underbrace{\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}}_B u(t)$$
$$y(t) = \underbrace{\begin{bmatrix} c_1 & c_2 \end{bmatrix}}_C \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

State-space representation

General ($n \geq 2$):

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), & x(0) &= x_0 \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

where we have introduced the *state vector* as

$$x(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

which contains all the internal knowledge necessary to determine the future behavior of the system. Here, $x(0) = x_0$ represents the initial state (for $t = 0$).

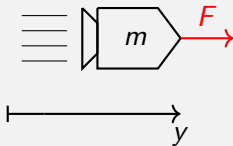
Note: usually, $D = 0$.

To go from a differential equation to a state-space model:

- I. Physical states [Exempel 8.2, 8.3]
- II. Diagonal form [Exempel 8.5]
- III. Controllable canonical form [Exempel 8.6]
- IV. Observable canonical form [Exempel 8.7]

From differential equations to state-space model

Example (I. Physical states)



Control signal: $u = F$, power/force

Output signal: y , position

$$\{ \text{Newton II} \} \implies m\ddot{y} = F \implies \ddot{y}(t) = \frac{1}{m}u(t) \quad (\text{Double integrator})$$

$$\text{Now let } \begin{cases} x_1(t) &= y(t) & \text{position} \\ x_2(t) &= \dot{y}(t) & \text{velocity/speed} \end{cases}$$

$$\implies \begin{cases} \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \end{cases}$$

In a DC motor, the transfer function is given by

and the output signal is then obtained by

which gives us

16


$$\begin{cases} \dot{x}_1(t) &= u(t) \\ \dot{x}_2(t) &= -x_2(t) - u(t) \\ y(t) &= x_1(t) + x_2(t) \end{cases}$$
$$A = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

17

From state-space form to transfer function

$$\left. \begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned} \right\} \xRightarrow{\mathcal{L}} \begin{aligned} sX(s) &= AX(s) + BU(s) \\ Y(s) &= CX(s) + DU(s) \end{aligned}$$

Note, however, that we have assumed that $x(0) = 0$!

$$\Rightarrow (sI - A)X(s) = BU(s) \Rightarrow X(s) = (sI - A)^{-1}BU(s)$$

and

$$Y(s) = CX(s) + DU(s) = \left(C(sI - A)^{-1}B + D \right) U(s)$$

i.e.

$$G(s) = D + C(sI - A)^{-1}B$$

Given the system

we calculate

which we can most easily invert using the rule for the inverse of 2×2 matrices

19



In our example, that means

$$\implies C(sI - A)^{-1}B = \frac{1}{s^2}$$

This gives a double integrator!

20

Poles of the system = eigenvalues of A

The **poles of the system** are given by:

$$\det(sI - A) \stackrel{\text{in the example}}{=} s^2 \quad (\text{characteristic polynomial})^*$$

Note that this gives the *eigenvalues* of A :

$$Ax = \lambda x \Rightarrow (\lambda I - A)x = 0 \Rightarrow \det(\lambda I - A) = 0$$

In general, the following applies

$$G(s) = \frac{C(sI - A)^* B}{\det(sI - A)} + D$$

where $(sI - A)^*$ is the adjoint matrix of $(sI - A)$.

poles of the system = eigenvalues of the matrix A



A system $\dot{x}(t) = Ax(t)$ ($x(0) = x_0$) is *asymptotically stable* if $\lim_{t \rightarrow \infty} x(t) = 0$ for all initial conditions $\bar{x}$.

A linear system is asymptotically stable if and only if all the eigenvalues of the matrix A are in the LHP (have strictly negative real part)

A linear system is input-output (I/O) stable if all the eigenvalues of the matrix A are in the LHP (have strictly negative real part)

22



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Example (Nonlinear system)

$$\begin{cases} \dot{x} &= x + x^2 + \sin u \\ y &= e^x + \cos u \end{cases}$$


$$\begin{aligned} x(t) &\rightarrow \bar{x}, \quad t \rightarrow \infty \\ (\dot{x}(t) &\rightarrow 0, \quad t \rightarrow \infty) \end{aligned}$$
$$\begin{aligned} f(\bar{x}, \bar{u}) &= 0 \\ h(\bar{x}, \bar{u}) &= \bar{y} \end{aligned}$$
$$\begin{aligned}\Delta x &= x - \bar{x} \\ \Delta u &= u - \bar{u} \\ \Delta y &= y - \bar{y}\end{aligned}$$

Linearization

The derivative of Δx can be calculated as:

$$\begin{aligned}\Delta \dot{x} &= \frac{d}{dt}(x - \bar{x}) = \dot{x} = f(\bar{x} + \Delta x, \bar{u} + \Delta u) = \{ \text{Taylor} \} = \\ &= \underbrace{f(\bar{x}, \bar{u})}_{=0} + f_x(\bar{x}, \bar{u})\Delta x + f_u(\bar{x}, \bar{u})\Delta u + \text{higher-order terms (h.o.t.)}\end{aligned}$$

Observe that we are dealing with matrices here:

$$\begin{aligned}f(x, u) &= \begin{bmatrix} f_1(x_1, x_2, \dots, x_n, u) \\ \vdots \\ f_n(x_1, x_2, \dots, x_n, u) \end{bmatrix} \\ \Rightarrow f_x &= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \nabla f_1 \\ \vdots \\ \nabla f_n \end{bmatrix}\end{aligned}$$

Similarly, for Δy we obtain

$$\begin{aligned}\Delta y &= y - \bar{y} = h(\bar{x} + \Delta x, \bar{u} + \Delta u) - \bar{y} = \{ \text{Taylor} \} = \\ &= \underbrace{h(\bar{x}, \bar{u}) - \bar{y}}_{=0} + h_x(\bar{x}, \bar{u})\Delta x + h_u(\bar{x}, \bar{u})\Delta u + \text{h.o.t.}\end{aligned}$$

This gives us the expressions for the *linear approximation*:

$$\begin{aligned}\Delta \dot{x} &= A\Delta x + B\Delta u \\ \Delta y &= C\Delta x + D\Delta u\end{aligned}$$

where

$$\begin{aligned} A &= f_x(\bar{x}, \bar{u}), & B &= f_u(\bar{x}, \bar{u}) \\ C &= h_x(\bar{x}, \bar{u}), & D &= h_u(\bar{x}, \bar{u}) \end{aligned}$$

Important: This only applies to small Δx , Δu , and Δy !

Question Can one determine whether the nonlinear system is stable near a steady state by analyzing the corresponding linearized system?

Answer: Yes, the nonlinear system is locally asymptotically stable if the poles of the linear system lie strictly in the left half-plane.

This result is called *Lyapunov's theorem* and is proven using Grönwall's inequality (Swedish-born mathematician):

If $f(t)$ is continuous and

$$f(t) \leq \alpha + \beta \int_0^t f(s) ds, \quad \forall t > 0$$

where $\alpha \leq 0$ and $\beta \leq 0$, then $f(t) \leq \alpha e^{\beta t}$, $\forall t > 0$

Note that $\frac{d}{dt}f(t) \leq \beta f(t)$, $f(0) \leq \alpha$